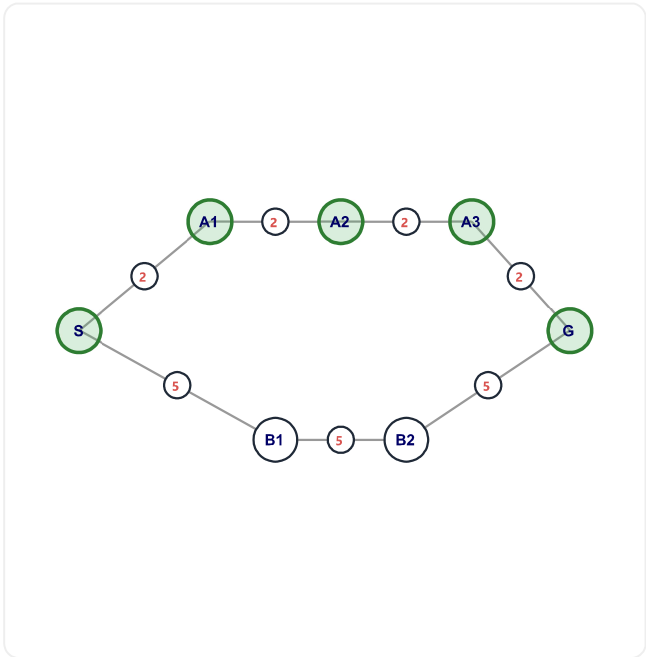


Select Lab Module:

Part 1: Calculation Lab (Greedy vs A*)

Reset Module

The Map & Edge Costs: g(n)



Heuristic Table: h(n)

Node	S	A1	A2	A3	B1	B2	G
h(n)	8	6	4	2	4	2	0

1. Greedy Best-First
2. A* Search

A* Search

Formula: $f(n) = g(n) + h(n)$

Path: S → A1 → A2 → A3 → G

Unexpanded Nodes (Frontier):
Goal Reached!

Calculate the Next Expansion

Node Name	g(n) Cost	h(n) Heuristic
E.G. A1	0	0
f(n) Total		0

Evaluate & Expand

Correct! Expanded G.
 $f(G) = 8 + 0 = 8$.
🏆 SEARCH COMPLETE!

A* found the Optimal Path! (Cost: 8)

Even though B1 looked closer ($h=4$), A* calculated the true past cost **$g(n)=5$** . The total f-score for B1 was 9, while the A-route maintained a perfect f-score of 8. A* safely navigated the backroads!